

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 30

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Class Test

1. Define Big Data and explain why traditional data processing approaches are often insufficient for big data environments.
2. Explain the 5Vs of Big Data with one practical example for any two Vs.
3. Differentiate structured, semi-structured, and unstructured data with examples.
4. Explain the role of distributed file systems in big data processing.
5. List any three major security concerns in big data platforms and briefly explain each.
6. What are data appliances and integration tools? Explain their role in big data workflows.
7. Discuss two major challenges organizations face while implementing big data analytics.
8. Write short notes on any one domain application of big data analytics.
9. How do business drivers influence the adoption of big data solutions?
10. Explain the importance of big data analytics in decision making.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage

Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

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ASSESSMENT TOOL 1: Class Test — Questions with Answers

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Assessment Tasks — Class Test (Answers)

Question 1

Define Big Data and explain why traditional data processing approaches are often insufficient for big data environments.

Answer

Definition

Big Data refers to data assets whose **volume, velocity and variety** are so large that conventional single-machine databases and tools can no longer capture, store, manage and analyse them within an acceptable time and cost. It is defined not by an absolute size but by the point at which **traditional technology fails**, and it therefore demands distributed storage, parallel processing and horizontally scalable architectures.

Why traditional approaches fall short

Aspect	Traditional Approach (RDBMS / DWH)	Why It Is Insufficient for Big Data
Scaling model	Vertical scaling — buy a bigger server (more CPU/RAM)	Hardware has a physical and cost ceiling; petabyte workloads need horizontal scale-out across commodity nodes
Schema	Schema-on-write; rigid, fixed schema defined before loading	JSON logs, images, video and sensor feeds have no fixed schema; schema-on-read is required
Data type support	Optimised for structured rows and columns	Roughly 80–90% of enterprise data is semi/unstructured and cannot be modelled as tables
Processing speed	Batch ETL, nightly loads, single-node execution	Clickstream and IoT data arrive continuously; insight is needed in seconds, not next morning

Fault tolerance	Depends on backups and expensive HA hardware	In a 1000-node cluster node failure is routine; replication and automatic re-execution are mandatory
Cost	Licensed, proprietary, high cost per TB	Commodity hardware plus open-source (Hadoop, Spark) or cloud object storage is far cheaper per TB
Data movement	Move data to the compute/processing server	Moving petabytes saturates the network; big data moves <i>computation to the data</i> (data locality)

Conclusion: Traditional systems fail on the three axes of **scale, structure and speed** simultaneously. Big data frameworks (HDFS, Spark, Kafka, NoSQL) succeed because they distribute both storage and computation, tolerate failure by design and defer schema enforcement until read time.

Question 2

Explain the 5Vs of Big Data with one practical example for any two Vs.

Answer

V	Meaning	Typical Measure	Technology Implication
Volume	Sheer quantity of data generated and stored	TB / PB / EB per day	Distributed storage (HDFS, S3), partitioning, compression
Velocity	Speed at which data arrives and must be processed	Events per second; latency	Stream processing (Kafka, Spark Streaming, Flink)
Variety	Different formats and sources of data	Structured / semi / unstructured	Data lake, NoSQL, schema-on-read
Veracity	Trustworthiness, accuracy and noise level in data	% missing, duplicate, inconsistent	Data cleansing, validation rules, lineage, master data management
Value	Usable business benefit extracted from the data	ROI, uplift, cost saved	Analytics, ML models, dashboards, decision support

Practical Example 1 — Volume (Retail / E-commerce)

- A large online retailer such as Amazon or Flipkart records **billions of clickstream events and millions of orders daily**, easily crossing several petabytes per year.
- No single database server can hold or scan this; the data is split into blocks and stored across hundreds of nodes in **HDFS / cloud object storage**.
- Queries run in parallel on each node (MapReduce / Spark), so a scan that would take days on one machine finishes in minutes.

Practical Example 2 — Velocity (Credit Card Fraud Detection)

- A payment network processes **tens of thousands of card swipes per second**, and each transaction must be approved or declined within roughly **100–300 milliseconds**.
- Transactions stream through **Apache Kafka** and are scored in-flight by a model running on **Spark Streaming / Flink**, with recent customer behaviour cached in Redis.
- Here the challenge is not storing the data but **reacting to it before the transaction completes** — a pure velocity problem.

Key point: The 5Vs are not independent — an architecture decision taken for one V (e.g. streaming for velocity) constrains the others (storage format for volume, governance for veracity). **Value** is the only V the business actually pays for; the other four are the costs incurred to reach it.

Question 3

Differentiate structured, semi-structured, and unstructured data with examples.

Answer

Basis	Structured	Semi-structured	Unstructured
Definition	Data organised in a fixed, predefined schema of rows and columns	Data with tags/markers giving partial organisation, but no rigid schema	Data with no predefined model or internal organisation
Schema	Schema-on-write (fixed before load)	Self-describing; schema can evolve per record	No schema at all
Examples	Bank transaction tables, ERP records, sales tables, sensor readings in a CSV	JSON API responses, XML, log files, e-mails with headers, HTML pages	Videos, photographs, X-ray / MRI images, audio calls, free-text reviews, PDFs
Storage	RDBMS (MySQL, Oracle), columnar warehouses (Redshift, BigQuery)	Document / NoSQL stores (MongoDB, Elasticsearch), Parquet, Avro	Object storage / data lake (HDFS, Amazon S3), blob stores
Query method	Direct SQL	SQL-on-JSON, XPath, Hive/Spark with nested types	Requires ML, NLP, computer vision or manual tagging to become queryable
Share of enterprise data	Approx. 10–20%	Approx. 10–15%	Approx. 70–80%
Processing difficulty	Low	Medium	High

Illustration — one hospital record, three data types

- **Structured:** patient ID, age, blood pressure reading stored in the hospital EHR table.
- **Semi-structured:** the insurance claim submitted as an XML/JSON document with optional and repeating fields.
- **Unstructured:** the MRI scan image and the doctor's dictated consultation note.

Analysis: A modern big data platform must handle all three *together*. This is why the **data lake** pattern (raw storage first, schema applied at read time) replaced the traditional warehouse-only design — a warehouse can only accept the first column of the table above.

Question 4

Explain the role of distributed file systems in big data processing.

Answer

A **Distributed File System (DFS)** such as HDFS, GFS or a cloud object store spreads one logical file across many physical machines, while presenting it to the user as a single file. It is the storage foundation on which all big data processing frameworks sit.

Core roles performed

Role	How It Works	Benefit to Big Data Processing
Horizontal scalability	Files split into fixed blocks (e.g. 128 MB) and spread over many DataNodes; new nodes added on demand	Petabyte-scale storage on cheap commodity hardware

Fault tolerance	Each block replicated (default 3 copies) on different nodes/racks; NameNode re-replicates on failure	Node or disk failure causes no data loss and no job failure
Data locality	The scheduler sends the computation task to the node that already holds the block	Avoids network bottleneck — "move computation, not data"
Parallel throughput	Hundreds of disks read simultaneously	Aggregate bandwidth in GB/s instead of MB/s from one disk
Write-once, read-many model	Files appended rather than randomly updated	Simplifies consistency and suits analytical scan workloads
Format flexibility	Stores any bytes — CSV, Parquet, JSON, video	Enables the data lake and schema-on-read
Foundation for frameworks	MapReduce, Hive, Spark, HBase all read/write through the DFS API	One storage layer serves batch, SQL, ML and streaming engines

HDFS architecture in brief

Client → **NameNode** (master: keeps metadata, block map, replication policy) → returns block locations → **Client** reads/writes directly from **DataNodes** (slaves: store actual blocks, send heartbeats + block reports)

- The NameNode never carries the data itself, only the metadata — this keeps the master lightweight.
- Rack-awareness places replicas on at least two racks so a whole rack failure is survivable.
- Loss of heartbeats triggers automatic re-replication, so the cluster self-heals.

Conclusion: Without a DFS there is no big data processing — it converts a fleet of unreliable, ordinary machines into one reliable, enormous, high-throughput storage device, which is precisely the substrate parallel analytics needs.

Question 5

List any three major security concerns in big data platforms and briefly explain each.

Answer

#	Security Concern	Explanation	Typical Countermeasure
1	Data privacy & unauthorised access	A data lake concentrates personal, financial and health data from many sources in one place, creating a single high-value target. Combining harmless datasets can also re-identify individuals who were anonymous in each source alone.	Role-based / attribute-based access control (Apache Ranger, Sentry), Kerberos authentication, masking and tokenisation of PII, k-anonymity or differential privacy
2	Weak encryption of data at rest and in transit	Data is replicated across hundreds of nodes and moves constantly over the network. Default Hadoop configurations historically shipped with authentication and encryption disabled , so a stolen disk or a sniffed link exposes raw records.	HDFS Transparent Data Encryption with encryption zones, TLS/SSL for RPC and shuffle traffic, KMS-managed key rotation, disk-level encryption
3	Insufficient auditing, monitoring and lineage	With hundreds of users and automated jobs, it becomes difficult to prove <i>who</i> accessed <i>what</i> and <i>when</i> . Without audit trails, insider misuse and slow data exfiltration go undetected, and regulatory compliance (GDPR, HIPAA, PCI-DSS) cannot be demonstrated.	Centralised audit logs, SIEM integration, data lineage tools (Apache Atlas), anomaly detection on access patterns, immutable log storage

Additional concerns (brief)

- **Distributed attack surface:** every node, API and open port in the cluster is an entry point; misconfigured cloud storage buckets are a common breach cause.
- **Insecure NoSQL stores:** many NoSQL databases lack mature built-in authentication and are vulnerable to injection through query documents.
- **Endpoint / source validation:** spoofed IoT devices or fake events can poison the dataset and mislead the models trained on it.

Analysis: Big data security is harder than conventional security because controls must hold across **many nodes, many formats and many tools** at once. Security therefore has to be designed into the pipeline (encrypt, authenticate, authorise, audit at every hop) rather than bolted on at the perimeter.

Question 6

What are data appliances and integration tools? Explain their role in big data workflows.

Answer

Data Appliances

A **data appliance** is a pre-integrated, purpose-built package of hardware plus software (server, storage, network, database and analytics engine) delivered by one vendor and optimised end-to-end for analytical workloads. Examples: **Oracle Exadata, Teradata, IBM Netezza (PureData), SAP HANA appliance**, and cloud equivalents such as Amazon Redshift or Snowflake offered as managed "appliance-like" services.

- Ships tuned and ready — little installation, sizing or performance tuning effort required.
- Uses massively parallel processing (MPP), columnar storage and often in-memory or FPGA-based filtering.
- Single point of support and predictable performance, but higher cost and vendor lock-in.

Data Integration Tools

Integration tools move, combine, cleanse and transform data from many heterogeneous sources into a common analytical store. Categories and examples:

Category	Representative Tools	Function in the Workflow
Batch ingestion / DB import	Apache Sqoop, AWS DMS, Informatica	Bulk transfer between RDBMS and HDFS/warehouse
Log & file ingestion	Apache Flume, Fluentd, Logstash, NiFi	Collect and route server, application and device logs
Streaming / message bus	Apache Kafka, Amazon Kinesis, Pulsar	Decouple producers from consumers; buffer high-velocity events
ETL / ELT & transformation	Talend, Informatica, Apache Spark, dbt	Clean, join, standardise, aggregate and enrich data
Workflow orchestration	Apache Airflow, Oozie, Luigi	Schedule, sequence, retry and monitor pipeline stages
Metadata & quality	Apache Atlas, Great Expectations	Catalogue, lineage, validation and governance

Their role in the big data workflow

Sources (OLTP DB, logs, IoT, APIs, social) → **Integration tools** (Kafka / Sqoop / NiFi) → **Data lake & processing** (HDFS + Spark) → **Data appliance / warehouse** (curated, high-speed query layer) → **BI dashboards & ML models**

- **Integration tools solve the Variety problem** — they are the plumbing that makes scattered, mismatched sources usable together.
- **Appliances solve the performance problem** — they serve the curated, structured layer to analysts at interactive speed.

- Together they shorten time-to-insight, reduce hand-written code, and enforce consistency and quality across the pipeline.

Trend: Physical appliances are steadily being replaced by **cloud-native, elastically scaled services** (Snowflake, BigQuery, Databricks), which give appliance-like tuning without the capital cost or fixed capacity.

Question 7

Discuss two major challenges organizations face while implementing big data analytics.

Answer

Challenge 1 — Data quality, integration and governance ("garbage in, garbage out")

- Data arrives from dozens of systems with **different formats, keys, units, time zones and definitions**; the same customer may exist five times with five IDs.
- Duplicate, missing, stale and inconsistent records directly corrupt models and dashboards — an inaccurate insight is worse than no insight because it is acted upon.
- Ungoverned lakes degrade into **"data swamps"**: data nobody can find, trust, or explain the origin of.
- **Mitigation:** master data management, a data catalogue with lineage (Apache Atlas), automated validation rules at ingestion, clear data ownership/stewardship, and documented single definitions of business metrics.

Challenge 2 — Skill shortage and high cost of talent and infrastructure

- A working platform needs data engineers, data scientists, DevOps/cloud engineers and domain experts — a scarce and expensive combination.
- Cluster licensing, cloud compute and storage costs grow quickly, and **ROI is not immediate**, so management support can lapse before value appears.
- Many projects stall at the proof-of-concept stage because there is no team to productionise, monitor and maintain the pipelines.
- **Mitigation:** start with one narrowly scoped, high-value use case; use managed cloud services to cut operational burden; upskill existing staff; adopt self-service BI so analysts do not depend on engineers for every query.

Comparison of the two challenges

Dimension	Data Quality & Governance	Skills & Cost
Nature	Technical + organisational	Human + financial
Symptom	Conflicting reports, unreliable models, data swamp	Stalled pilots, over-run budgets, unmaintained pipelines
Business impact	Wrong decisions taken with confidence	No decisions taken at all; project abandoned
Primary remedy	Governance framework, catalogue, validation	Phased scope, managed services, training

Other notable challenges: security and regulatory compliance (GDPR/HIPAA), scalability and real-time latency demands, integration with legacy systems, and organisational resistance to data-driven decision making.

Question 8

Write short notes on any one domain application of big data analytics.

Answer

Domain chosen: Healthcare Analytics

Healthcare generates enormous, highly varied and highly sensitive data — electronic health records, laboratory results, medical imaging, genomic sequences, wearable device streams, insurance claims and clinical notes. Big data analytics converts this into better outcomes at lower cost.

Data sources and their nature

Source	Data Type	Analytical Use
Electronic Health Records (EHR)	Structured	Risk scoring, readmission prediction, cohort analysis
Insurance claims	Semi-structured (XML/JSON)	Fraud detection, cost analysis, utilisation review
Medical imaging (MRI, CT, X-ray)	Unstructured	Deep-learning based tumour and anomaly detection
Wearables & ICU monitors	Streaming time-series	Real-time deterioration alerts, remote patient monitoring
Genomic sequences	Unstructured, very large	Precision medicine, drug response prediction
Clinical notes	Unstructured text	NLP extraction of symptoms, adverse drug events

Key applications

- **Predictive diagnosis and early warning:** models flag sepsis or cardiac deterioration hours before clinical signs appear.
- **Personalised / precision medicine:** treatment tailored to a patient's genomic profile and history rather than population averages.
- **Hospital operations:** forecasting admissions to plan beds, staff rosters and drug inventory.
- **Epidemic and public-health surveillance:** aggregating case and mobility data to track outbreak spread.
- **Claims fraud detection:** spotting improbable billing patterns across providers.
- **Drug discovery:** mining trial and molecular data to shorten development cycles.

Benefits vs. challenges

Benefits	Challenges
Earlier detection and better clinical outcomes	Strict privacy law (HIPAA / GDPR) — heavy consent and audit burden
Reduced cost through fewer readmissions and less waste	Data siloed across hospitals with no common standard
Efficient use of beds, staff and equipment	Model bias and the "black box" problem in clinical trust
Population-level insight for public health policy	Very high veracity requirement — an error can cost a life

Analysis: Healthcare is the domain where **Veracity and security dominate over Volume**. The technical pipeline resembles any other big data stack, but the governance, consent, de-identification and explainability layers are far heavier.

Question 9

How do business drivers influence the adoption of big data solutions?

Answer

Business drivers are the commercial pressures and opportunities that make an organisation invest in big data. Technology is adopted *because* of a driver, and the driver determines which architecture is actually chosen.

Business Driver	Influence on Adoption	Resulting Technical Choice
Competitive pressure	Rivals already personalise and price dynamically; not adopting means losing share	Recommendation engines, ML platforms, A/B testing infrastructure

Customer experience & retention	Need for a 360° single view of the customer across all channels	Customer data platform, identity resolution, churn models
Cost reduction & efficiency	Cheaper commodity/cloud storage vs. expensive licensed warehouses	Data lake on object storage, open-source stack, elastic cloud scaling
Revenue growth / new products	Data itself monetised or used to launch data-driven services	APIs, analytics-as-a-service, real-time personalisation
Risk & fraud management	Losses from fraud justify sub-second detection spend	Streaming platform (Kafka + Flink), anomaly detection models
Regulatory compliance	Mandatory reporting, audit and data-retention obligations	Lineage, immutable audit logs, encryption, governance tooling
Operational agility / speed	Decisions needed in real time, not from a monthly report	Real-time dashboards, stream processing, in-memory stores
Technology maturity & cloud availability	Low entry barrier — pay-as-you-go removes capital risk	Managed services (EMR, Databricks, BigQuery)

Nature of the influence

- **They set priority and budget:** a fraud-loss driver funds a low-latency streaming stack; a compliance driver funds governance tooling instead.
- **They define success metrics:** ROI is measured against the driver (churn %, fraud loss, cost per TB), not against technical benchmarks.
- **They determine urgency and scope:** a regulatory deadline forces adoption regardless of readiness; a cost driver allows a slower, phased rollout.
- **They secure executive sponsorship:** projects tied to a clear business driver survive budget review; purely technical projects usually do not.

Conclusion: Big data adoption is **business-pull, not technology-push**. Organisations that begin from a specific business driver deliver measurable value; those that begin from "we should have a Hadoop cluster" typically produce an expensive, unused data swamp.

Question 10

Explain the importance of big data analytics in decision making.

Answer

Big data analytics shifts organisations from **intuition-based decisions** to **evidence-based decisions**, and from decisions taken monthly on samples to decisions taken continuously on complete data.

The four analytics levels supporting decisions

Level	Question Answered	Technique	Decision Example
Descriptive	What happened?	Reporting, dashboards, aggregation	Sales fell 8% in the south region last quarter
Diagnostic	Why did it happen?	Drill-down, correlation, root-cause analysis	The fall traces to stock-outs of two SKUs
Predictive	What will happen?	Machine learning, forecasting, regression	These 12,000 customers are likely to churn next month
Prescriptive	What should we do?	Optimisation, simulation, recommendation engines	Offer this specific discount to these customers now

Why it matters — key points

- **Accuracy:** decisions rest on full populations rather than small samples, cutting guesswork and bias.
- **Speed:** real-time streams enable decisions within seconds — approve a payment, reroute a driver, throttle a fraudulent session.
- **Personalisation:** supports decisions at the level of the individual customer instead of a broad segment.
- **Risk reduction:** early detection of fraud, equipment failure or credit default prevents loss instead of reporting it afterwards.
- **Cost optimisation:** demand forecasting reduces inventory, staffing and energy waste.
- **Innovation:** patterns discovered in data reveal unmet needs and new product opportunities.
- **Measurability:** A/B testing lets each decision be validated with evidence rather than opinion.

Before and after comparison

Aspect	Traditional Decision Making	Big Data–Driven Decision Making
Basis	Experience, intuition, sample surveys	Complete, current, multi-source data
Latency	Weekly or monthly reports	Real-time / near real-time
Granularity	Segment or region level	Individual customer, device or transaction
Orientation	Reactive — explains the past	Proactive — predicts and prescribes
Validation	Rarely measured after the fact	Continuously tested and refined

Caution: Analytics supports but does not replace judgement. Poor data quality, biased training data or misread correlation as causation can make a confident decision a wrong one — so **governance and human oversight remain essential**.

— End of Assessment Tool 1 (Class Test) —